Xiaocong Yang

Ph.D. Student, CS @ UIUC | Founder, AI Interpretability @ Illinois

Personal Website | Blog Posts | Google Scholar | LinkedIn | xy51@illinois.edu

Education

Ph.D. in Computer Science | University of Illinois Urbana-Champaign **2025.8 - 2029.5 (expected)**

- Founder and current leader of *AI Interpretability @ Illinois* research group.

M.S. in Computer Science | University of Illinois Urbana-Champaign **2023.8 - 2025.5**

B.S. in Economics | Tsinghua University **2018.9 - 2023.6**

Leadership & Teaching

Founder & Group Leader | AI Interpretability @ Illinois **2025.8 - present**

- Founded and lead UIUC's cross-lab research group on AI interpretability, with 4 faculty advisors, 40+ student researchers, and 5 concurrent research projects at this moment with bi-weekly advising meetings.
- Secured 45k+ GPU-hours on National Center for Supercomputing Applications (NCSA) Delta through ACCESS program as the PI, and $20k+ computing credits from Microsoft Azure and AWS.
- Established external collaborations with faculties and Ph.D. students from top peer institutions.

Instructor | CS 591 Biologically Plausible AI, UIUC **Spring 2026, Fall 2026**

- *Independently delivered 2 class sessions on AI Interpretability in the Era of LLMs: Architecture, Behavior and Beyond* [Section 1][Section 2]

Instructor | CS 591 Biologically Plausible AI, UIUC **Fall 2025**

- *Independently delivered 4 class sessions on A Tutorial on Interpretable and Biologically Plausible LLM* [Section 1][Section 2][Section 3][Section 4]

Publication

13.Xiaocong Yang*, Yuren Hao*, ChengXiang Zhai. DAGFormer: Global Topology Prediction for Transformer Cross-Layer Connections. In review of ARR, [link]

12.Xiaocong Yang, Xuanyu Meng, Yuji Zhang, Jiawei Han, ChengXiang Zhai. Sparse Autoencoders Are a Shared Coordinate System for Symbolic Matching. In review of ARR, [link]

11. Ali Ebrahimpour-Boroojeny*, **Xiaocong Yang***, Samuel Schapiro, Zahra Borhani, Francisco R Ortega, Yujia Zheng, ChengXiang Zhai. Resisting Harmful Instructions through Training-Free Weight Editing. In review of ARR, [link]

10.Xiaocong Yang*, Rishit Chatterjee*, Himanshu Udupi, Vikram Ramavarapu, ChengXiang Zhai. Skip the Compressor: Coding Agents Fix More Issues When They Pick Whole Files. In review of ARR, [link]

9. Xiaocong Yang. Generative Interpretability via Scalable Neuro-Symbolic Models, **ACM AI Leadership Summit 2026** [link]

8. Himanshu Udupi, **Xiaocong Yang**, ChengXiang Zhai. Globally Optimal Training of Spiking Neural Networks via Parameter Reconstruction, in submission to AISTATS 2027 [link]

7. Xiaocong Yang. LLM2IR: Simple Unsupervised Contrastive Learning Makes Long-Context LLM Great Retriever. M.S. Thesis [link]

6. Xiaocong Yang, Jiacheng Lin, Ziqi Wang, Chengxiang Zhai. *Learning by Analogy: Enhancing Few-Shot Prompting for Math Word Problem Solving with Computational Graph-Based Retrieval*, preprint [link]

5. Ziyi Chen, **Xiaocong Yang**, Jiacheng Lin, Chenkai Sun, Kevin Chen-Chuan Chang, Jie Huang. *Cascade Speculative Drafting for Even Faster LLM Inference*, **NeurIPS 2024** [link]

4. Xiaocong Yang, James Huang, Wenxuan Zhou, Muhao Chen. *Parameter-efficient Tuning with Special Token Adaptation*, **EACL 2023** [link]

3. Xingcheng Yao, Yanan Zheng, **Xiaocong Yang**, Zhilin Yang. *NLP from Scratch Without Large-Scale Pretraining: A Simple and Efficient Framework,* **ICML 2022** [link]

2. Leixian Shen, Enya Shen, Yuyu Luo, **Xiaocong Yang**, Xuming Hu, Xiongshuai Zhang, Zhiwei Tai, Jianmin Wang. *Towards natural language interfaces for data visualization: A survey*, **TVCG 2022** [link]

1. Hao Zhou, Pei Ke, Zheng Zhang, Yuxian Gu, Yinhe Zheng, Chujie Zheng, Yida Wang, Chen Henry Wu, Hao Sun, **Xiaocong Yang**, Bosi Wen, Xiaoyan Zhu, Minlie Huang, Jie Tang. *Eva: An open-domain chinese dialogue system with large-scale generative pre-training*, preprint [link]

Research

Research Assistant | **TIMAN Group, University of Illinois Urbana-Champaign** **2024.3 - present**
Advisor: Chengxiang Zhai | *Donald Biggar Willett Professor, UIUC* Urbana-Champaign

- Proposed and published several papers ranging from AI Interpretability, Biologically Plausible AI and Information Retrieval (publication #6 - #13).
- Co-instructed CS591 Biologically Plausible AI starting from Fall 2025 and continuing.
- Founded and led AI Interpretability @ Illinois research group under his supervision.

Research Assistant | **FORWARD Data Lab, University of Illinois Urbana-Champaign** **2023.8 - 2024.1**
Advisor: Kevin Chenchuan Chang | *Professor, UIUC* Urbana-Champaign

- Proposed *Cascade Speculative Drafting*, where we vertically and horizontally cascade multiple draft models of different size. Our approach is mathematically proved to be more efficient than standard speculative decoding and achieves up to **81%** additional speedup in the experiments.
- Paper is published at NeurIPS 2024.

Visiting Researcher | **LUKA Group, University of Southern California** **2022.4 - 2022.9**
Advisor: Muhao Chen | *Assistant Professor, Information Science Institute, USC* Los Angeles

- Proposed and led research on *PASTA*, a model finetuning method that is more efficient than all of existing baselines which tunes only **0.029%** of total parameters to match the performance of full finetuning.
- Paper is published at EACL 2023.

Research Assistant | **IIIS, Tsinghua University** **2021.7 - 2023.1**
Advisor: Zhilin Yang | *Assistant Professor, Institute for Interdisciplinary Information Sciences, Tsinghua University* | *Founder & CEO, Moonshot AI* Beijing

- Contributed to *Task-driven Language Modeling*, proposing a task-specific pretraining framework that reduced by **100x** the total training cost and size of corpus and democratizing the development of large LLMs.
- Analyzed the functionality of “vertical attention heads” in LLMs.
- Paper is published at ICML 2022.

Research Assistant | **Conversational AI (CoAI) Group** **2021.1 - 2021.6**
Advisor: Huang Minlie | *Associate Professor* | *Department of Computer Science and Technology, Tsinghua University* Beijing

- Contributed to the development of the open-domain dialogue system *EVA* and the corresponding paper.
- Designed the interface and client-server communication framework for the deployment of the dialog system, tested rule-based and model-based methods to filter out toxic content in the generated text.

Internship

Applied Research Scientist | **Pinterest** **2026.5-2026.8**
Supervisor: Weijie (Jenny) Jiang | *Senior Machine Learning Engineer, Pinterest Core Engineering*

- Proposed and implemented to apply Multi-gate Mixture-of-Experts (MMoE) architecture to the ranking model used in the recommendation system at Pinterest, and obtained up to **3.0%** improvement on multiple tasks including click and save compared to baseline model.

AI Scientist Intern | **Intuit** **2025.5-2025.8**
Supervisor: Ruocheng Guo | *Staff Research Scientist, Intuit AI Research* Mountain View

- Raised RestBench tool-call success by **+5 percentage points** by designing tool description improvement pipeline for agentic LLM flows; scoped and led end-to-end study at Intuit.

Invited Talk

Guest Speaker | **Citadel GQS PhD Colloquium** **2026.4**
New York

- Invited to represent AI Interpretability @ Illinois and give a presentation about generative interpretability of LLMs and our current work to Citadel's core quantitative research teams.